A STRICT HALL INEQUALITY FOR THE AUXILIARY BIPARTITE GRAPH OF THE GALLAI-EDMONDS DECOMPOSITION

SAHAR DISKIN, ANNA GEISLER

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Allowed standard prerequisites (editorial).

- The bipartite graph Q between the set A and the components D_i of the Gallai-Edmonds decomposition, and the maximum matchings M_v leaving a vertex v of D unmatched. Set up in the author's subsection on the Edmonds-Gallai decomposition (source lines 1054-1072), together with the author's displayed Tutte-Berge and Gallai-Edmonds relations. Supplies the objects that the strict Hall inequality is proved for, and the family of maximum matchings that the proof inspects.

Lemma 0.1. *For every non-empty $S \subseteq A$,*

$$|N_Q(S)| \geq |S| + 1.$$

Consequently every vertex of A has neighbours in at least two components of $G_p[D]$, and Q has at most r connected components.

Proof. For every i and every $v \in D_i$, the definition of D gives a maximum matching M_v leaving v unmatched; in particular, in M_v there is no edge from A to D_i. The edges in M_v match A to distinct members of $\{D_j : j \neq i\}$. Thus $Q - \{D_i\}$ has a matching saturating A. In particular, Q has a matching saturating A, so Hall's theorem gives $|N_Q(S)| \geq |S|$. If $|N_Q(S)| = |S|$, deleting any $D_i \in N_Q(S)$ would violate Hall's condition, a contradiction.

In each component of Q, the number of components of $G_p[D]$ is therefore at least one more than the number of vertices in A. Summed over all components of Q, this difference is $m - a = r$, so Q has at most r components. □

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